

## **Three Common Social Engineering Attacks and Mitigation Strategies**

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#### **Scareware**

Fear is a common tactic to encourage an innocent victim into providing personal information or money to an attacker. Scareware exploits the fear tactic pushing the victim into purchasing fake software disguised as some type of cybersecurity software like anti-virus or anti-malware. The scareware typically prompts the victim with pop-up notifications in their browser or on a webpage notifying the user that dangerous files were detected on the device and require the purchase of security software to resolve the issue. Cybercriminals receive money by the purchase of the software from the unsuspecting user which is harmless bloatware but can go a step further by adding malware to the scareware software the user downloads (Gillis, 2021).

By adding the initial tactic of scaring the user into believing there is malicious software on the device, the attacker is capable of getting the user to actually download such malicious software. Then the attacker has legitimate access to spy on, steal data, demand ransom for locked files, and attempt to deceive the victim further in many other ways to gain more information and money. Bloatware can cause a user's system to slow down and can potentially make the user more susceptible to other attacks. The user or organization will have lost money on the software but will typically not be affected further. However, if the bloatware downloaded does contain malware, serious consequences can arise depending on the type of malware and other security controls the user or organization already have implemented (Gillis, 2021).

The easiest way to defend systems against scareware is to have legitimate anti-virus installed on all endpoints on the network. Known scareware that has been detected can be removed by trusted software. Other mitigation strategies include keeping browsers updated, implementing firewalls, and informing employees not to click on or provide information to pop-up windows or untrusted websites asking for personal information (Gillis, 2021).

#### **Watering Hole**

Attackers who seek to target a specific group might use a watering hole attack which requires an attacker to determine common websites users of a particular group frequent often. The attacker will inject malicious code into the site triggering a payload when users visit the site. The sites that are typically targeted are messaging boards or general interest sites. A site that has been maliciously altered is difficult to detect and can be destructive if it is able to breach layers of security controls or target third-party vendors of the intended group. However, the websites targeted generally have little to no security making a watering hole attack easy to implement (Shacklett et al., 2025).

Once the payload is executed, the attacker can gain access to various other assets in the organization and on its network with the ability to cause significant damage to an organization in the form of data breaches, reputational harm, or installing malware. To prevent watering hole attacks, organizations need to follow best practices for modern cybersecurity and proper hardening of devices including installing anti-virus software and the use of web proxies and web logging to monitor online activity of employees. Awareness training should discourage

employees from ignoring security warnings and clicking questionable links on websites (Shacklett et al., 2025).

### **Honeypot**

On the side of the organization, honeypots can be used to lure in unsuspecting malicious actors. This decoy system is protected with several layers of security though configured to maintain the appearance of being a vulnerable and attractive system for attackers. Researchers and specialists will implement honeypots to track any attempts at accessing the system to understand the level and type of threats the organization is facing while simultaneously distracting potential attackers from the functional systems. Honeypots tend to be placed inside a demilitarized zone to keep them isolated from the internal network and prevent hackers from accessing legitimate private information (Robinson et al., 2025).

While they can be used to collect useful intelligence, honeypots are expensive and difficult to maintain requiring specialists to implement and monitor them closely. Hackers may be able to hijack an improperly implemented honeypot to gather information about the organization as a form of reconnaissance. Honeypots, combined with other modern cybersecurity technologies like next-generation firewalls and intrusion detection systems, can help researchers and specialists gain practical insights into the threat landscape and understand how they can improve on their on-going cybersecurity practices (Robinson et al., 2025).

### **References**

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